

End-to-End Machine Learning Pipeline Assignment using MySQL and Streamlit:

Titanic Dataset

[<https://drive.google.com/file/d/1qXYOq8RnR5IkOTpNMGkUEqEwB9ccsWQo/view?usp=sharing>]

MySQL Titanic Data:

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with 'titanic' selected. The 'Columns' tab for the 'titanic' table is active, showing the following columns: PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, and Embarked. The 'Result Grid' on the right displays a sample of data from the 'titanic' table.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	S	S
2	1	1	Cunings, Mrs. John Bradley (Florence Briggs Th...)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heldunen, Miss. Laina	female	26	0	0	STON/O2, 3101282	7.925	S	S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05	S	S
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	E46	S
8	0	3	Palsson, Master. Gosta Leonard	male	2	3	1	349909	21.075	S	S
9	1	3	Johnson, Mrs. Oscar W (Elizabeth Vilhelmina Berg)	female	27	0	2	247742	11.1333	S	S
10	1	2	Nasser, Mrs. Nicholas (Adele Adhem)	female	14	1	0	237726	30.0708	C	S
11	1	3	Sandstrom, Miss. Marguerite Rut	female	4	1	1	PP 9549	16.7	G6	S
12	1	1	Bonnell, Miss. Elizabeth	female	58	0	0	113783	26.55	C103	S
13	0	3	Soundercock, Mr. William Henry	male	20	0	0	A/5 2151	8.05	S	S
14	0	3	Anderson, Mr. Anders Johan	male	39	1	5	347082	31.275	S	S
15	0	3	Vestrom, Mrs. Hilda Amanda Adolfsa	female	14	0	0	350406	7.8542	S	S
16	1	2	Hewlett, Mrs. (Mary D Kingcome)	female	55	0	0	248706	16	S	S
17	0	3	Rice, Master. Eugene	male	2	4	1	382652	29.125	Q	S
19	0	3	Vander Planke, Mrs. Julius (Emelia Maria Vande...)	female	31	1	0	345763	18	S	S
21	0	2	Fynney, Mr. Joseph J	male	35	0	0	239865	26	S	S

Streamlit APP(titanic.py):

Random Forest:

The screenshot shows a web browser window displaying the 'Titanic Survival Prediction App'. The app interface is dark-themed and contains several input fields for user data. The 'Select Model' dropdown is set to 'Random Forest'. The input fields are: Passenger Class (3), Sex (female), Age (27.00), Number of Siblings/Spouses Aboard (2), Number of Parents/Children Aboard (2), Fare (500.00), and Embarked (S). A 'Predict' button is located below the input fields. The prediction result, 'The passenger did not survive.', is displayed at the bottom of the form. The browser's address bar shows the URL 'titanic-app-6djh7z4i58cxjrlz7r8nt.streamlit.app'. The Windows taskbar at the bottom shows the time as 12:50 AM on 5/2/2025.

Titanic Survival Prediction App

Select Model
Random Forest

Passenger Class
3

Sex
female

Age
27.00

Number of Siblings/Spouses Aboard
2

Number of Parents/Children Aboard
2

Fare
500.00

Embarked
S

Predict

The passenger did not survive.

XGBoost:

The screenshot shows the same 'Titanic Survival Prediction App' interface, but with the 'Select Model' dropdown set to 'XGBoost'. The input fields are: Passenger Class (1), Sex (female), Age (35.00), Number of Siblings/Spouses Aboard (1), Number of Parents/Children Aboard (2), Fare (15000.00), and Embarked (S). The 'Predict' button is highlighted in red. The prediction result, 'The passenger survived!', is displayed at the bottom of the form. The browser's address bar and the Windows taskbar (showing 12:53 AM on 5/2/2025) are identical to the previous screenshot.

Titanic Survival Prediction App

Select Model
XGBoost

Passenger Class
1

Sex
female

Age
35.00

Number of Siblings/Spouses Aboard
1

Number of Parents/Children Aboard
2

Fare
15000.00

Embarked
S

Predict

The passenger survived!